

DL4DS Midpoint Report: Fine-Grained Visual Classification of Bird Species

Shiheng Xu, Renjie Fan, Kunshu Yang

April 2025

Abstract

This midpoint report provides an update on our project exploring Fine-Grained Visual Classification (FGVC) using deep learning. We focus on the CUB-200-2011 dataset and have trained a ResNet50 baseline. We also performed exploratory data analysis and structured our dataset pipeline. Our next steps include integrating Swin Transformers, attention mechanisms, and supervised contrastive learning to improve performance and interpretability.

Introduction

Fine-Grained Visual Classification (FGVC) remains a challenging problem in computer vision due to subtle inter-class differences and large intra-class variability. Since our original proposal, we have refined our goal to not only build accurate classifiers but also systematically compare CNN-based and Transformer-based architectures with emphasis on interpretability and robustness. Our updated approach combines empirical evaluation with visualization-based analysis.

Related Work

FGVC has traditionally been addressed using convolutional neural networks with transfer learning. ResNet50 and similar architectures serve as strong baselines. More recent research has incorporated attention mechanisms such as CBAM and bilinear pooling, and introduced Transformer models like Swin Transformer, which leverage hierarchical self-attention. We are exploring relevant work on Swin Transformer[5], supervised contrastive learning, and attention heatmap techniques such as Grad-CAM.

Dataset and Exploratory Analysis

We use the CUB-200-2011 dataset, which consists of 11,788 images across 200 bird species. Using the official split files, we parsed the dataset and loaded images with their associated labels. All images were resized to 224x224 resolution and normalized using ImageNet preprocessing statistics. We conducted exploratory data analysis to examine the class distribution, image sizes, and overall data quality. The dataset was found to be relatively balanced across categories, and the average image size was around 400x300 pixels. Visual inspection of randomly sampled images further confirmed the consistency and appropriateness of the data for our task.

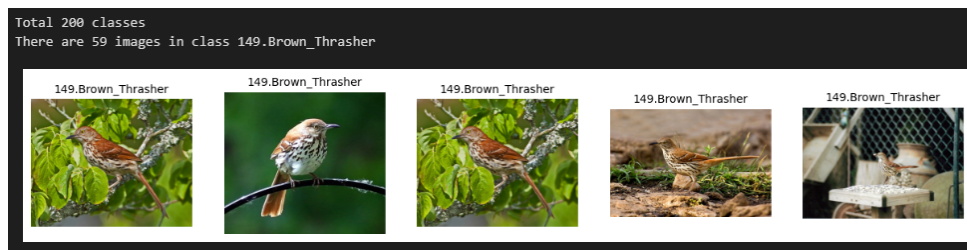


Figure 1: Example visualizations of class 149 (Brown Thrasher). The dataset consists of 200 total classes.

Preliminary Experiments

We implemented a baseline model using ResNet50 with pretrained ImageNet weights and trained it using PyTorch. Our pipeline includes standard data augmentation techniques, cross-entropy loss, and SGD optimization with momentum. The training was conducted on GPU with a batch size of 16 over 10 epochs.

This baseline model achieved progressive improvements in test accuracy across epochs, starting from approximately 41% and reaching a final test accuracy of 73.4%. This demonstrates the model’s strong capacity to learn meaningful fine-grained features even without specialized architectural enhancements.

Updated Plan and Timeline

By Week 9, we have completed the baseline model training, exploratory data analysis, and dataset processing pipeline setup. In the following weeks (10–11), we plan to integrate and train a Swin Transformer model on the same dataset split to compare its performance with ResNet50. Weeks 12–13 will focus on incorporating Grad-CAM for interpretability analysis and exploring supervised contrastive learning as a possible alternative loss function. In the

```

Using device: cuda
C:\Users\27435\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.11_qbz5n2k...
warnings.warn(
C:\Users\27435\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.11_qbz5n2k...
warnings.warn(msg)
Epoch 1/10: 100%|██████████| 188/188 [00:18<00:00, 9.91it/s]
Epoch 1/10, Loss: 4.0108
Test Accuracy: 0.4147
Epoch 2/10: 100%|██████████| 188/188 [00:17<00:00, 10.52it/s]
Epoch 2/10, Loss: 1.9382
Test Accuracy: 0.6137
Epoch 3/10: 100%|██████████| 188/188 [00:18<00:00, 10.32it/s]
Epoch 3/10, Loss: 1.0581
Test Accuracy: 0.6593
Epoch 4/10: 100%|██████████| 188/188 [00:18<00:00, 9.91it/s]
Epoch 4/10, Loss: 0.6308
Test Accuracy: 0.6805
Epoch 5/10: 100%|██████████| 188/188 [00:18<00:00, 10.38it/s]
Epoch 5/10, Loss: 0.3882
Test Accuracy: 0.6943
Epoch 6/10: 100%|██████████| 188/188 [00:18<00:00, 10.23it/s]
Epoch 6/10, Loss: 0.2370
Test Accuracy: 0.6995
Epoch 7/10: 100%|██████████| 188/188 [00:18<00:00, 10.40it/s]
Epoch 7/10, Loss: 0.1664
Test Accuracy: 0.7244
Epoch 8/10: 100%|██████████| 188/188 [00:18<00:00, 10.29it/s]
Epoch 8/10, Loss: 0.1098
Test Accuracy: 0.7209
Epoch 9/10: 100%|██████████| 188/188 [00:18<00:00, 10.33it/s]
Epoch 9/10, Loss: 0.0690
Test Accuracy: 0.7325
Epoch 10/10: 100%|██████████| 188/188 [00:17<00:00, 10.45it/s]
Epoch 10/10, Loss: 0.0503
Test Accuracy: 0.7344
Training complete!

```

Figure 2: Training log showing epoch-wise loss and test accuracy for ResNet50 baseline.

final week, we will finalize model comparisons, conduct error analysis, and prepare the project report and presentation deliverables.

Conclusion

At this midpoint, we have successfully established a functional baseline and validated the dataset pipeline. Our work so far confirms the tractability of the task and sets a foundation for further experimentation. Moving forward, we aim to enhance model capacity, incorporate interpretability techniques, and evaluate the comparative strengths of CNNs and Transformers in the fine-grained classification context.

References

- [1] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.
- [2] Tsung-Yu Lin, Aruni RoyChowdhury, and Subhransu Maji, "Bilinear CNN models for fine-grained visual recognition," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2015.
- [3] Sanghyun Woo, Jongchan Park, Joon-Young Lee, and In So Kweon, "CBAM: Convolutional Block Attention Module," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018.
- [4] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.
- [5] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo, "Swin Transformer: Hierarchical Vision Transformer using Shifted Windows," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.
- [6] Catherine Wah, Steve Branson, Peter Welinder, Pietro Perona, and Serge Belongie, "The Caltech-UCSD Birds-200-2011 Dataset," *Technical Report*, California Institute of Technology, 2011.
- [7] Olga Russakovsky et al., "ImageNet Large Scale Visual Recognition Challenge," *International Journal of Computer Vision (IJCV)*, vol. 115, no. 3, pp. 211–252, 2015.